

```

```text
haut : [01][02][03][04][05][06][07][08][09][10][11][12][13][14][15][16][17][18][19][20][21][22]
[23][24]
 | | |
 CLK GND A15
 | | | |
 /IRQ P1-2 P1-3 P1-4

bas : [48][47][46][45][44][43][42][41][40][39][38][37][36][35][34][33][32][31][30][29][28][27]
[26][25]
 | | | | |
 5V GND 02 01 00
```

```

Additional lines on the same safe proto bundle:

```
```text
`19 = PORT1-0`
`20 = PORT1-1`
```
```

So the **safe first-choice wiring set** is:

```
1. `48` = `+5V`
2. `47` = `GND`
3. `02` = `GND` secondaire
4. `45` = `OUT2`
5. `44` = `OUT1`
6. `43` = `OUT0`
7. `01` = `PORT1-CLK`
8. `19` = `PORT1-0`
9. `20` = `PORT1-1`
10. `15` = `PORT1-2`
11. `16` = `PORT1-3`
12. `18` = `PORT1-4`
13. `14` = `/IRQ` optionnel
14. `05` = `A15` bonus debug
```

With a `16-wire ribbon`, the last `2` conductors stay available for:

```
15. spare / reserve
16. spare / reserve
```

Advanced bundle (more powerful, more risky)

This is the other direction we discussed, but **not** the recommended bring-up path:

```
- `CPU D0..D7`
- `A15`
- `/IRQ`
- later `R/W` and `M2`
```

Keep this for a later bus-aware proto, not for proto 1.

Signal list

Top row

```
- `01` = `PORT1-CLK`
- `02` = `GND`
- `05` = `A15`
- `14` = `/IRQ` with `1 kOhm` series resistor
- `15` = `PORT1-2`
- `16` = `PORT1-3`
- `18` = `PORT1-4`
- `19` = `PORT1-0`
- `20` = `PORT1-1`
```

Bottom row

```
- `48` = `+5V`
- `47` = `GND`
- `45` = `OUT2`
- `44` = `OUT1`
- `43` = `OUT0`
- `32` = `CPU D0`
- `31` = `CPU D1`
- `30` = `CPU D2`
- `29` = `CPU D3`
```

- `28` = `CPU D4`
- `27` = `CPU D5`
- `26` = `CPU D6`
- `25` = `CPU D7`

Quick reality check before soldering all lines

- `02` and `47` should both be `GND`
- `43`, `44`, `45` should be the three contiguous `OUT0`, `OUT1`, `OUT2` lines
- in this **proto 1 safe** version, `01` is used as `PORT1-CLK`, not as a power pin

If that matches your meter, the orientation is locked.

Cable / connector note

For a first V2 bundle, we need roughly `16` lines.

What exists in standard parts:

- `16-conductor ribbon cable`
- `HE10 / IDC 16-pin connectors`
- `16-pin IDC board headers`

What probably does **not** exist as a neat off-the-shelf "perfect V2 cable":

- a ready-made round cable already terminated exactly for this project

So the realistic options are:

1. flat `16-way ribbon` with IDC/HE10 ends
2. ribbon cable that is separated for a short distance near the motherboard
3. custom round multicore cable only if we later want a cleaner final build

Sources

- https://www.nesdev.org/wiki/Expansion_port
- <https://commons.wikimedia.org/wiki/File:Nintendo-NES-Mk1-Motherboard-Bottom.jpg>
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